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Class: I5-AMS-DAS-B

1. Introduction

It purposely provides a background on Python programming practices and why writing structured, maintainable code is essential for software development

- **Function:** A block of code that performs a specific task.
- **Variable:** A name used to store data.
- **Method:** A function that belongs to a class.
- **Class:** A blueprint for creating objects.
- **Module:** A Python file that contains reusable code.
- **Package:** A collection of related Python modules.
- **Constant:** A value that we intend not to change.

2. Clean Code

Clean code means that to writing code that is easy to read, understand, and maintain by both yourself and other developers. Clean code focuses on clear variable names, concise logic, and modular design.

I also explained to the whole class about this section as well.

3. PEP 8

PEP 8 is a style guide that helps us write clean and readable Python code.

- **Indentation:** Use 4 spaces for each indentation level.
- **Line length:** Keep lines reasonably short, usually no more than 79 characters.
- **Imports:** Put import statements at the beginning of the file.
- **Naming conventions:** Use snake_case for variables and functions, UPPER_CASE for constants, and PascalCase for classes.
- **Whitespace:** Use spaces properly around operators and after commas to make code easier to read.

4. Debugging

Debugging means finding and fixing errors in our code. When an error occurs, we should read the error message, check the code, identify the problem, and test the code again.

5. Tips and Tricks to Help Ensure Code Follows PEP 8

- Use meaningful variable and function names.
- Avoid repeating the same code unnecessarily.
- Follow PEP 8 rules.
- Add comments when they are helpful.
- Avoid hardcoding values that may need to change.
- Test the code regularly.
- Use debugging to find and fix errors.
- Keep the code simple and easy to understand.